

Author Response: “When Does Conformer Geometry Help? Selective Complementarity of 3D Ensemble Statistics and 2D Fingerprints”

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We thank all three reviewers for their thoughtful and constructive feedback. Below we address each point in detail. All changes in the revised manuscript are marked in [blue text](#); line references (e.g., [L42--44]) refer to the revised `main.tex`.

Reviewer 1P2E (Rating: 4, Borderline Accept)

R1.1: Supplement empirical findings with a theoretical discussion explaining why simple covariance invariants outperform more complex representations (bias-variance trade-off, parameterization complexity, dataset size). **Response:** We have substantially expanded the “simplicity principle” discussion in Section 6 [L615--618]. The revised text [L618] now provides an explicit parameter-count analysis: the 5 scalar invariants operate at ~ 180 samples per covariance parameter on ESOL ($N_{\text{train}} \approx 900$), whereas the lowrank variant ($k=128$) operates at ~ 7 —well below the threshold for reliable estimation. We connect this to classical model selection theory, showing that the bias-variance tradeoff penalizes higher-dimensional representations at the sample sizes typical of molecular property datasets (642–4,200 molecules). The new feature dimension ablation (Table 6, Section 5.6, [L491--515]) provides independent confirmation: even for the mean μ , RMSE monotonically degrades from $D=256$ to $D=2,048$, demonstrating that this is a fundamental statistical constraint rather than an engineering limitation.

R1.2: Conduct supplementary experiments on classification benchmarks. **Response:** We have added classification experiments on BACE (1,513 molecules, β -secretase inhibition) and BBBP (2,038 molecules, blood-brain barrier permeability) in a new Section 5.5 [L468--483]. Results (Table 7, [L475]) confirm the property taxonomy extends to classification: conformer features hurt BACE by -4.4% AUROC and show no benefit on BBBP ($+0.1\%$). This is consistent with the taxonomy— β -secretase binding is an electronic/binding property, and BBB permeability is governed primarily by topological features. Per-seed results are provided in Appendix K [L1056--1079].

R1.3: Validate key findings on larger or industrially sourced datasets. **Response:** We acknowledge this as an important limitation in the revised Conclusion [L635--636]. Our benchmarks span 642 to 133K molecules across academic datasets (MoleculeNet, QM9, MARCEL); validation on larger industrially sourced datasets (ChEMBL, ZINC, proprietary ADMET panels) would strengthen the taxonomy’s generalizability. We note this as a priority for future work and have added explicit language to this effect in Section 7 (Limitations, [L636]).

R1.4: Increase figure resolution, standardize metrics, correct typographical errors.

Response: We have conducted a full proofreading pass. All figures are exported at ≥ 300 DPI. Evaluation metrics are standardized (RMSE throughout for regression, AUROC for classification, [L468--483]). Typographical errors and formatting inconsistencies have been corrected across tables and main text (see blue-marked changes throughout).

R1.5: Provide clearer descriptions of scaffold split generation and number of replicates.

Response: We have expanded Section 3.7 (Training Details, [L281]) with explicit descriptions of the Murcko scaffold split procedure (Bemis–Murcko generic scaffolds extracted via RDKit, sorted by frequency, assigned to splits maintaining 80/10/10 ratios) and seed counts (3 seeds for the main benchmark, 10 seeds for statistical validation experiments). See the new blue text at [L281].

Reviewer tcvY (Rating: 3, Borderline Reject)

R2.1: Insufficient 3D GNN baselines—only SchNet is used; need DimeNet++, PaiNN, Equiformer.

Response: We have added PaiNN [Schütt et al., 2021] as a second 3D GNN baseline (Section 5.4, Table 5 at [L441], Appendix L at [L1081--1103]). PaiNN uses equivariant message passing with separate scalar/vector channels—a fundamentally different inductive bias from SchNet’s continuous-filter convolutions. Results confirm the hierarchy: PaiNN achieves 28% (ESOL), 42% (FreeSolv), and 30% (Lipophilicity) gains over FP+XGBoost from a *single* conformer, while SchNet uses 10.

The convergence of 21–42% gains across these two architecturally diverse GNNs provides strong evidence that the hierarchy reflects a genuine property of the representation, not an artifact of SchNet specifically. We explain in the new paragraph at [L460] why DimeNet++ (intermediate between SchNet and PaiNN in design) and Equiformer (primarily beneficial in large-scale pre-training settings orthogonal to our study) are not included: two architectures spanning the continuous-filter and equivariant paradigms are sufficient to establish robustness.

Regarding the reviewer’s concern that SchNet could be trivially extended with ensemble pooling: this is precisely what our DKO mean ensemble baseline does (mean pooling over conformers through a neural architecture, [L354]). The fact that even this approach underperforms FP+XGBoost strengthens, rather than undermines, the claim that the pre-computed feature bottleneck is the limiting factor.

R2.2: FreeSolv results are directionally inconsistent (−13.5% to +18.0%).

Response: We have substantially clarified this in the revised scaffold split discussion at [L573]. The +18.0% degradation comes from a 3-seed random-split auxiliary experiment on $N=642$ molecules—a regime where a single unfavorable split can dominate the mean. The definitive result is the 10-seed paired one-sided t -test on the main scaffold-split pipeline: −13.5% improvement, $p < 3 \times 10^{-5}$ (Table 4, [L525]). The learning curve analysis (Section 5.7, [L575--578]) further clarifies: hybrid features hurt at small data fractions but become beneficial above ~ 440 molecules, explaining the sensitivity to split and seed variation at FreeSolv’s small sample size. The 10-seed result, not the 3-seed auxiliary, should be taken as definitive.

R2.3: Feature extraction pipeline (zero-padding/truncation to fixed D) may be the primary bottleneck.

Response: We have added a new discussion paragraph at [L515] addressing this concern directly. The feature dimension ablation (Table 6, [L498]) demonstrates that RMSE *monotonically degrades* from $D=256$ to $D=2,048$. If truncation were the bottleneck, we would expect larger D to help—instead, it hurts. This shows the limitation is the curse of

dimensionality (noise dimensions that shallow models cannot exploit), not information loss from truncation. We acknowledge that BDE’s variable dimensions (187–1,854) are a genuine concern and note adaptive-length encoding as future work at [L515], while emphasizing that this would not address the fundamental finding that pre-computed summary statistics discard the relational structure preserved by end-to-end GNNs.

R2.4: Excluding pre-trained models limits generalizability of the taxonomy.

Response: We agree that pre-trained 3D models (e.g., Uni-Mol) may alter the taxonomy boundaries. We have expanded the Related Work (Section 2, [L176]) and Limitations (Section 7, [L636]) to explicitly scope the taxonomy to the non-pre-trained setting and acknowledge this limitation. The new text at [L176] states: “This scoping decision means our property taxonomy is validated in the non-pre-trained setting; pre-trained 3D models such as Uni-Mol may extract useful 3D signals even for electronic properties through learned representations trained on millions of conformers, which could alter the taxonomy boundaries.” Importantly, our DKO conformer statistics are *complementary* to pre-trained representations—they could be concatenated with Uni-Mol embeddings just as they are concatenated with fingerprints here. Testing this combination is an important direction for future work, but is orthogonal to our central question of when 3D conformer geometry *per se* adds predictive value beyond 2D fingerprints.

R2.5: Restriction to regression tasks limits the framework’s claimed generality.

Response: We have addressed this with new classification experiments on BACE and BBBP (Section 5.5, [L468--483]; Table 7, [L475]). Conformer features hurt BACE (−4.4% AUROC) and show no benefit on BBBP (+0.1%), consistent with the taxonomy (binding and permeability are not solvation-dependent). We have updated the scope statement in the Introduction ([L151]) and abstract ([L76]) to explicitly include classification targets. We note that neural DKO variants require architectural adaptation for classification (different loss function, output activation) and are evaluated only on regression; the hybrid FP+conformer approach is evaluated on both.

Reviewer u34v (Rating: 4, Borderline Accept)

R3.1: Manuscript feels too fragmented due to titled paragraphs.

Response: We have revised the Discussion (Section 6, [L594--624]) to reduce fragmentation. Bold titled paragraphs (“An empirical property taxonomy,” “Relationship to prior work,” “Practical implications”) have been converted to flowing prose with topic sentences that integrate the content more naturally. Specifically: the taxonomy opening at [L605] now uses a topic sentence rather than a bold header; the prior work transition at [L622] flows naturally from the preceding text; and the practical implications at [L624] open with a transitional phrase. The section now reads as a cohesive narrative rather than a series of independent blocks.

R3.2: Research questions formulated as actual questions feels too informal.

Response: We have rephrased RQ1–RQ3 in the Introduction ([L147--149]) as declarative objectives rather than interrogative questions. For example, RQ1 now reads “We characterize which molecular property types benefit from 3D conformer geometry and provide a mechanistic basis for the observed selectivity” rather than “Which molecular property types benefit from 3D conformer geometry, and what is the mechanistic basis for selectivity?”

R3.3: Performance hierarchy is confused and could benefit from clearer presentation.

Response: We have substantially rewritten the performance hierarchy presentation in Section 6

at [L613]. The revised text walks through each of the four tiers with explicit evidence and, crucially, explains the progression from tier 4 to tier 1 in terms of how much 3D relational structure is preserved: “DKO discards it into summary statistics, engineered descriptors preserve selected physical invariants, and GNNs retain the full atomic graph.” This mechanistic framing—absent from the original—makes the logic of the hierarchy explicit rather than presenting it as an empirical ranking without explanation. The key result—that FP+3D descriptors match SchNet on ESOL (1.000 vs. 1.004)—is highlighted as evidence that carefully engineered features can approach end-to-end learning.

R3.4: Some text too direct; expand into more technical discourse. **Response:** We have expanded terse statements throughout the manuscript into more technical language. Key locations: the abstract performance hierarchy ([L78]), the Discussion bias-variance analysis ([L618]), the hierarchy explanation ([L613]), the practical decision framework ([L624]), and the Conclusion ([L627--636]). These changes are all marked in blue.

Summary of changes (with line references to revised main.tex):

- Added PaiNN as second 3D GNN baseline [L434--460], Table 5 [L441], Appendix L [L1081--1103]
- Added classification benchmark on BACE and BBBP [L468--483], Table 7 [L475], Appendix K [L1056--1079]
- Added feature dimension ablation $D \in \{256, 512, 1024, 2048\}$ [L491--515], Table 6 [L498]
- Expanded bias-variance theoretical analysis [L615--618]
- Clarified FreeSolv directional inconsistency [L573]
- Addressed featurization bottleneck concern [L515]
- Expanded pre-training scoping and limitations [L176], [L636]
- Rewrote Discussion for clarity and flow [L605], [L613], [L622], [L624]
- Rephrased research questions as declarative objectives [L147--149]
- Clarified scaffold split procedure and replicate counts [L281]
- Added larger-dataset limitation acknowledgment [L636]
- All changes marked in [blue text](#)